FINAL REPORT

The Art of Scientific Computing: Respiratory Rates

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Chapter 1

Introduction

1.1 Respiration rate as an indicator of respiratory illness

The respiratory rate (the number of complete breathing cycles in one minute) is one the most fundamental indicators of respiratory illness due to there existing specific respiratory signals that indicate illness. Analysis of respiration rate is often the first step in diagnosing a respiratory illness, making respiratory rate an important physiological feature in treating specific illnesses.

Despite the importance of the respiratory rate, the process of analysing respiration is in general very difficult, as well as being time consuming and requiring specific training to be performed correctly. The fast-paced environment that healthcare workers are often exposed to makes it increasingly harder to avoid misdiagnosis through analysis of the respiratory rate. A study performed within two high-resource paediatric emergency departments in Western Australia found that between 6.9 to 14.5% of children were inappropriately prescribed treatment based on respiratory illness misdiagnosis [1]. This is a prevalent issue that is only perpetuated in low-resource settings, were respiratory illness misdiagnosis doesn't just serve as a danger to the patient, but can result in a further waste of resources through ineffective treatment.

An obvious solution to this problem is to have an automated process that identifies specific respiratory signals associated with respiratory illnesses, provided that those signals exist. This would help prevent misdiagnosis and the waste of resources, both supporting patient health and recovery, and lessening the strain upon low-resource healthcare facilities and workers.

[PARAGRAPH RELATING RESPIRATION SIGNALS TO ELECTROCARDIOGRAM SIGNALS]

This report focuses on the search for potential respiratory signals associated with obstructive sleep apnea (OSA) through ECG data. ...

1.2 Obstructive sleep apnea

1.3 The apnea-ECG database

Chapter 2

R-R peak interval

Chapter 3

QTc interval

Chapter 4

Conclusion

Bibliography

[1] P. Porter, J. Brisbane, J. Tan, N. Bear, J. Choveaux, P. Della, and U. Abeyratne. Diagnostic errors are common in acute pediatric respiratory disease: A prospective, single-blinded multicenter diagnostic accuracy study in australian emergency departments. *Frontiers in Pediatrics*, Volume 9 - 2021, 2021.